Fiscal Year:	FY 2018	**Task Last Updated:**	FY 10/01/2018
PI Name:	Snyder, Michael Ph.D.		
Project Title:	HERO Twin Astronaut Study Consortium (TASC) Project: Longitudinal integrated multi-omics analysis of the biomolecular effects of space travel		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94305-2200	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-TWINS
Start Date:	04/16/2014	**End Date:**	04/15/2018
No. of Post Docs:	4	**No. of PhD Degrees:**	2
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 4/15/2018 per PI and NSSC information (Ed., 4/21/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Winkelmann, Juliane M.D. (Stanford University)		
Grant/Contract No.:	NNX14AH52G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Key Objectives: The rigors of space travel present conditions for which our species have not adapted, and thus it would be a significant public health benefit to characterize physiologic and cellular responses to the unique environmental conditions associated with space travel including solar radiation, prolonged weightlessness, G-transitions, and circadian disturbances. Our goal is to perform a longitudinal integrative analysis of all omics data generated during the Human Exploration Research Opportunities (HERO) mission to provide an unprecedented portrait of the molecular and physiological changes that accompany space travel. Methods/Techniques: The Snyder laboratory at Stanford has recently pioneered longitudinal integrative personalized omics profiling (iPOP), which combines genome-wide sequencing with the regular sampling of a patient's transcriptome, proteome, metabolome, and cytokine profiles in order to detect multi-analyte signatures that correlate with disease onset and progression. Our approach will be to leverage our unique expertise in integrated omics studies to combine the data generated at each of the HERO sites in an integrated analysis framework to detect longitudinal multi-analyte signatures that originate specifically during or after the space mission. Using data-driven network inference, and knowledge-based pathway databases, we will build quantitative models of molecular network activity, observe temporal dynamics of network activity, and identify network modules associated with space travel. We will determine flight-specific biomolecules and pathways and detect direct, inverse, and phase-shifted correlations between flight-specific clinical and molecular signatures and significant events recorded during the space mission such as G-transitions during takeoff and landing, recorded solar events, etc. We are also collecting untargeted metabolomics and proteomics profiling data that will be a very useful complement to the other data types being collected, and will provide biological insights on physiology and metabolism associated with space flight. We hypothesize that through integrative omics analysis we will discover biological signatures associated with G-transitions, prolonged weightlessness, DNA damage response to solar radiation, and other environmental changes that occur during the space mission. The Snyder lab is also responsible for coordinating data transfer and storage by providing a secure environment for data storage and setting up secure data transfer methods.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed work is highly innovative and will provide the first large-scale omics data ever collected on a human during space travel and will present an unprecedented longitudinal analysis of monozygous twins placed in vastly different environments. This comprehensive biological analysis will allow us to better understand the health risks associated with space travel and develop strategies to mitigate these risks.
Task Progress:	Brief summary of research and results: To better understand the impact of spaceflight on the human body and to prepare for future exploration-class missions, the NASA Twins Study investigators leveraged the unique opportunity to study a pair of identical twin astronauts, before, during, and after NASA's first one-year mission. One twin (encoded TW) was onboard the International Space Station (ISS) while the other served as a genetically matched, ground control (encoded HR). Data were generated from specimens collected over 27 months before (preflight), during (inflight), and after (postflight) spaceflight using an integrated sampling scheme, spanning a wide variety of biological samples as well as cognitive and physiological measures. Longitudinal assessments included the genome, epigenome, transcriptome, proteome, metabolome, microbiome, and immunome, in addition to measures of molecular and physiological adaptations and cognitive function. A range of significant changes were observed during flight, including: (1) decreased body mass, (2) telomere elongation, (3) genome instability, (4) carotid artery distension and increased intima-media thickness, (5) altered ocular structure, (6) transcriptional and metabolic changes, (7) DNA methylation changes in several immune- and oxidative-stress related pathways, (8) changes in complex lipids, (9) shifts in composition and function of the gastrointestinal microbiota, and (10) cognitive performance decline. Upon return to Earth, the average telomere length and the majority of gene expression and microbiome changes observed during flight returned to near preflight levels; however, the expression levels of a subset of genes were still disrupted after six months back on Earth. Increased frequencies of chromosome structural variants were observed both during and after flight. Genome-wide DNA methylation changes were within normal ranges of variation, though locus-specific changes during flight were observed. Extended decreases in cognitive speed and accuracy were observed. Cytokines and other proteins involved in inflammatory, platelet, and cardiovascular responses were also disrupted after flight, indicating severe stress on the body upon return to gravity. Conclusion: Together, these data represent NASA's first multi-omics profiles of astronauts, NASA's first evaluation of the molecular, physiological, and cognitive effects of a year-long spaceflight mission, and the first such comprehensive comparison associated with spaceflight utilizing an Earth-bound twin as a control. Our study also pioneered new inflight data and sample collection strategies that were used to generate a comprehensive, longitudinal, molecular, physiological, and cognitive profile, thus providing a scientific framework and baseline data for future studies. As such, the data provide a valuable roadmap of the putative low-, medium-, and high-level risks for human spaceflight, which can help guide future biomedical measurements and countermeasures as astronauts embark on even longer missions to Mars and beyond.
Bibliography Type:	Description: (Last Updated: 08/12/2021)
Abstracts for Journals and Proceedings	The NASA Twins Investigators Group. "The NASA Twins Study: A multi-omic, molecular and physiological analysis of a year-long human spaceflight." To be presented at the 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018
Abstracts for Journals and Proceedings	Mishra T, Piening BD, Contrepois K, Ahadi S, Lee BA, Chen S, Salins D, Snyder MP. "Longtitudinal integrated multi-omics analysis of the biomolecular effects of space travel." 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018

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